Probability - 3.0.18

EE25BTECH11018 - Darisy Sreetej

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Question

Let $X_1, X_2, \cdots$ be a sequence of independent and identically distributed chi-square random variables, each having 4 degrees of freedom. Define

$$S_n = \sum_{i=1}^{n} X_i$$

If $\frac{S_n}{n} \to \mu$ as $n \to \infty$, then $\mu =$

1. 8
2. 16
3. 24
4. 32

(MA 2007)

Solution

A random variable $\mathbf{X}$ is said to follow a Chi-square distribution with n degrees of freedom if :

$$f(x) = \frac{1}{2^{n/2}\,\Gamma(n/2)}\, x^{\frac{n}{2}-1} e^{-x/2}, \quad x > 0 \tag{1}$$

$$f(x) = 0, \quad \text{otherwise} \tag{2}$$

$$\mathbf{X} \sim \chi^2(n) \tag{3}$$

(This is special case of Gamma Distribution)

Solution

$$\text{Given: } X \sim \chi^2(n), \quad f(x) = \frac{1}{2^{n/2}\,\Gamma(n/2)} x^{\frac{n}{2}-1} e^{-x/2},\ x > 0 \tag{4}$$

$$E[X] = \int_0^\infty x f(x)\, dx \tag{5}$$

$$= \int_0^\infty x \cdot \frac{1}{2^{n/2}\,\Gamma(n/2)} x^{\frac{n}{2}-1} e^{-x/2}\, dx \tag{6}$$

$$= \frac{1}{2^{n/2}\,\Gamma(n/2)} \int_0^\infty x^{\frac{n}{2}} e^{-x/2}\, dx \tag{7}$$

$$\text{Let } t = \frac{x}{2} \ \Rightarrow\ x = 2t,\ dx = 2\, dt \tag{8}$$

$$E[X] = \frac{1}{2^{n/2}\,\Gamma(n/2)} \int_0^\infty (2t)^{\frac{n}{2}} e^{-t} \cdot 2\, dt \tag{9}$$

Solution

$$= \frac{1}{2^{n/2}\,\Gamma(n/2)} \int_0^\infty 2^{n/2} t^{\frac{n}{2}} e^{-t} \cdot 2\, dt \tag{10}$$

$$= \frac{2^{n/2} \cdot 2}{2^{n/2}\,\Gamma(n/2)} \int_0^\infty t^{\frac{n}{2}} e^{-t}\, dt \tag{11}$$

$$= \frac{2}{\Gamma(n/2)} \int_0^\infty t^{\frac{n}{2}} e^{-t}\, dt \tag{12}$$

$$\text{Using Gamma function: } \int_0^\infty t^{m-1} e^{-t} dt = \Gamma(m) \tag{13}$$

$$\Rightarrow \int_0^\infty t^{\frac{n}{2}} e^{-t} dt = \Gamma\left(\frac{n}{2} + 1\right) \tag{14}$$

Solution

$$E[X] = \frac{2}{\Gamma(n/2)} \cdot \Gamma\left(\frac{n}{2} + 1\right) \tag{15}$$

$$\text{Using } \Gamma(m+1) = m\Gamma(m): \tag{16}$$

$$\Gamma\left(\frac{n}{2} + 1\right) = \frac{n}{2}\Gamma\left(\frac{n}{2}\right) \tag{17}$$

$$E[X] = \frac{2}{\Gamma(n/2)} \cdot \frac{n}{2}\Gamma\left(\frac{n}{2}\right) \tag{18}$$

$$= n \tag{19}$$

$$\boxed{E[X] = n} \tag{20}$$

Solution

According to our question , We have

$$X_1, X_2, \ldots \sim \chi^2(4) \quad \text{(independent and identically distributed)} \tag{21}$$

Also from (19) , $E[X_i] = 4$ Using **Law of Large Numbers(LLN)** , which states that the sample mean converges to the population mean as $n \to \infty$

$$\frac{1}{n}\sum_{i=1}^{n} X_i \to E[X_i] \quad \text{as } n \to \infty \tag{22}$$

Solution

Thus ,

$$\frac{S_n}{n} \to E[X_i] \quad \text{as } n \to \infty \tag{23}$$

$$\Rightarrow \frac{S_n}{n} \to 4 \quad \text{as } n \to \infty \tag{24}$$

Therefore , $\mu = 4$ (none of the options)

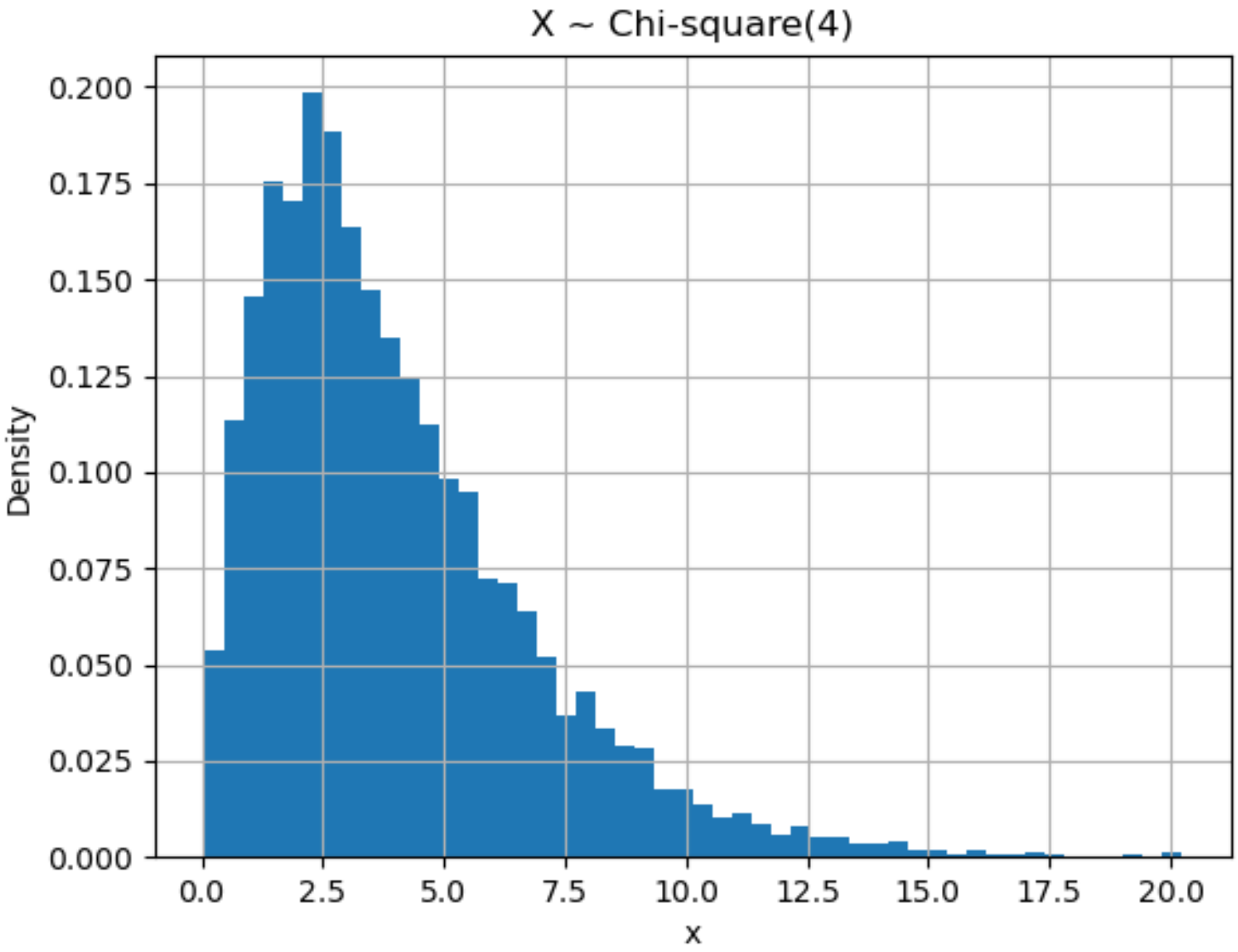
Python Code

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import chi2

# Parameters
n_max = 5000
df = 4

# Generate samples
X = np.random.chisquare(df, n_max)

#Plotting
plt.figure()
plt.hist(X, bins=50, density=True)
plt.title("X ~ Chi-square(4)")
plt.xlabel("x")
plt.ylabel("Density")
plt.grid(True)
```

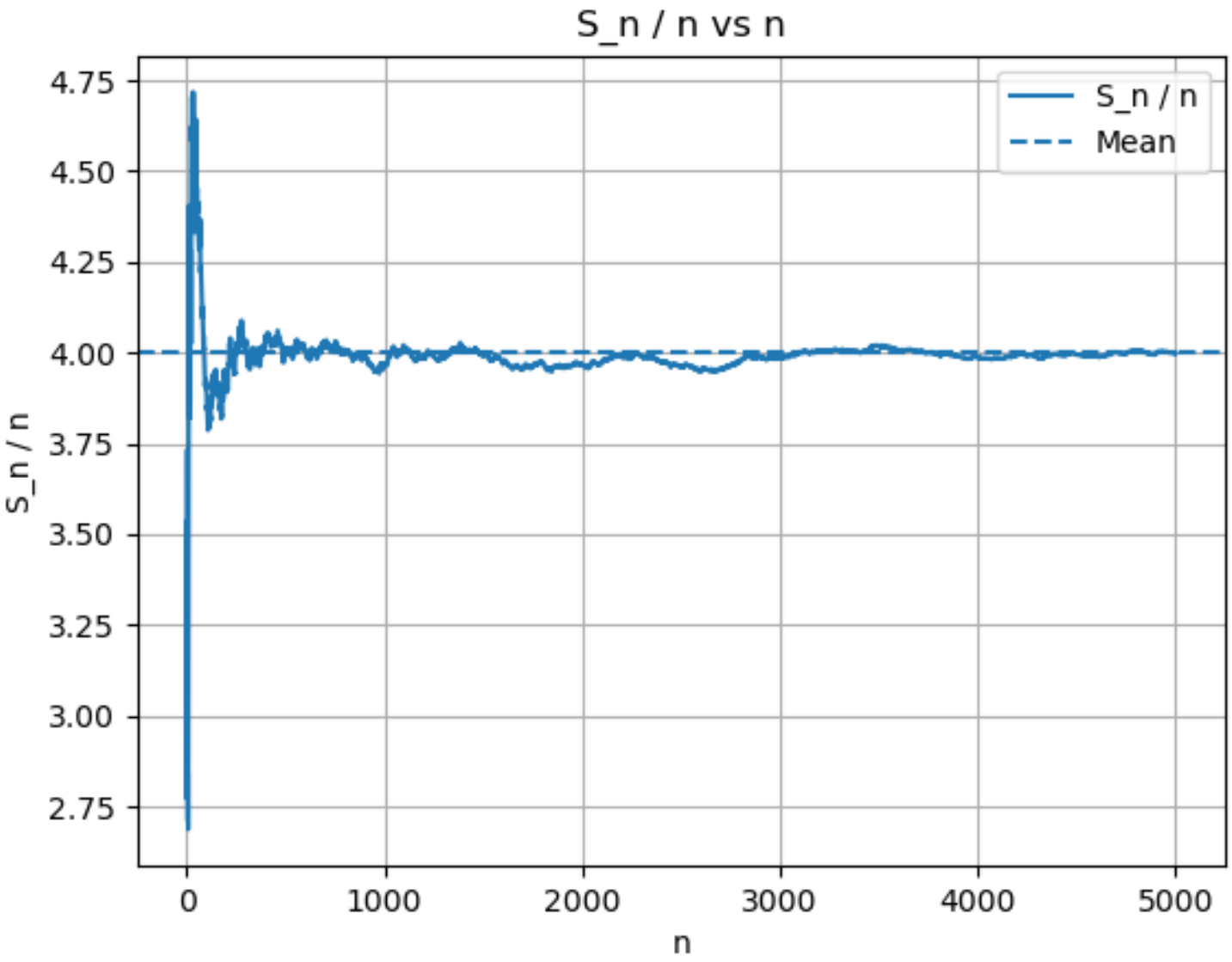
Python Code

```
S_n = np.cumsum(X)
n = np.arange(1, n_max + 1)
Sn_by_n = S_n / n

# Using scipy to get mean
true_mean = chi2.mean(df)

#Plotting
plt.figure()
plt.plot(n, Sn_by_n, label="S_n / n")
plt.axhline(y=true_mean, linestyle='--', label="Mean")
plt.title("S_n / n vs n")
plt.xlabel("n")
plt.ylabel("S_n / n")
plt.legend()
plt.grid(True)

plt.show()
```

Plot


X ~ Chi-square(4)
Density
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
x

Plot


S_n / n vs n
S_n / n
Mean
4.75
4.50
4.25
4.00
3.75
3.50
3.25
3.00
2.75
S_n / n
0
1000
2000
3000
4000
5000
n